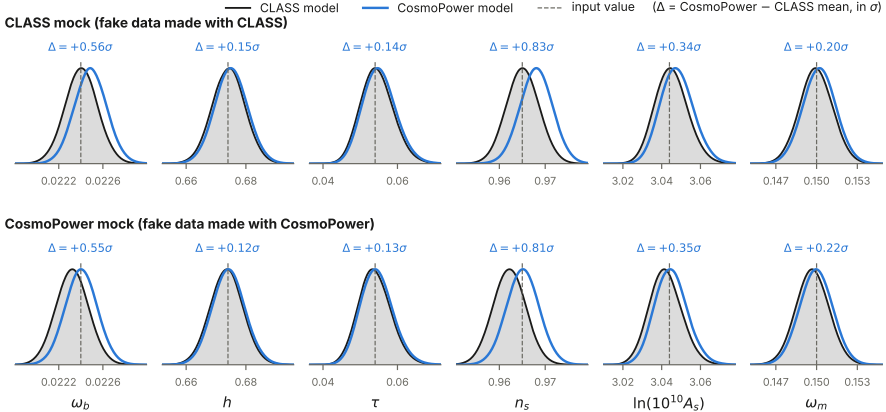
# How much does $\partial C_\ell/\partial\omega_m$ depend on the other parameters?

At the benchmark-1 point, $\partial C_\ell/\partial\omega_m$ recomputed with each ΛCDM parameter $\theta_i$ shifted by $\pm 3\%$; change measured with Planck-like noise weighting (33 $\mu$K-arcmin, 7′ beam, $f_{\rm sky} = 0.57$).

| | change in $\partial C_\ell/\partial\omega_m$, $\theta_i$ shifted $\pm 3\%$ | | | | same change at a |
|---|---|---|---|---|---|
| shifted $\theta_i$ | TT | TE | EE | $\phi\phi$ | $1\sigma$ shift (TT/TE/EE) |
| $h$ | 10.5% | 11.0% | 8.9% | 1.7% | $\sim 3\%$ |
| $\omega_b$ | 4.6% | 4.5% | 3.4% | 1.3% | $\sim 1\%$ |
| $A_s$ | 2.8% | 2.7% | 2.8% | 3.0% | $\sim 0.8\%$ |
| $n_s$ | 2.0% | 1.6% | 1.7% | 5.3% | $\sim 0.3\%$ |
| $\tau$ | 0.6% | 0.8% | 0.9% | 0.1% | $\sim 1.6\%$ |

For a $1\sigma$ shift in any other parameter, $\partial C_\ell/\partial\omega_m$ changes by at most $\sim 3\%$ (from $h$). A $\pm 3\%$ step is a much larger move than $1\sigma$ for $h$, $\omega_b$, $A_s$, $n_s$, and a smaller one for $\tau$, hence its larger $1\sigma$ value.

# Benchmark: CLASS vs CosmoPower, each fitted to both mocks




ΛCDM fit (one $\omega_m$ bin) to two mocks of the same cosmology, made with CLASS and with CosmoPower. 8 chains $\times 10^5$ steps per run.

Each code recovers its own mock ($|\text{pull}| \leq 0.12\sigma$). On either mock CosmoPower sits above CLASS by the same amount: $n_s$ $+0.8\sigma$, $\omega_b$ $+0.55\sigma$, $A_s$ $+0.35\sigma$, $\omega_m$ $+0.2\sigma$; $h$, $\tau \leq 0.15\sigma$. Errors identical.

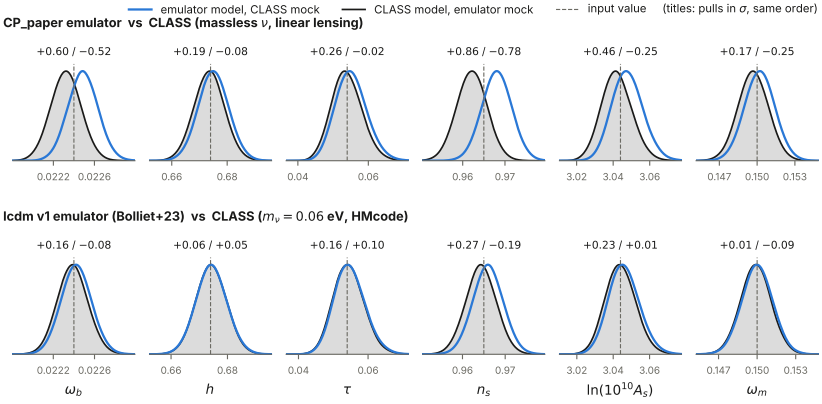
# Benchmark: numbers, and comparison with the Fisher forecast

| | CosmoPower − CLASS mean [$\sigma$] | | | error ratio | pull, code on its own mock [$\sigma$] | |
|---|---|---|---|---|---|---|
| | CLASS mock | CP mock | Fisher forecast | CP/CLASS | CLASS | CosmoPower |
| $\omega_b$ | $+0.56$ | $+0.55$ | $+0.55$ | 0.98 | $+0.03$ | $+0.03$ |
| $h$ | $+0.15$ | $+0.12$ | $+0.13$ | 1.00 | $+0.04$ | $+0.04$ |
| $\tau$ | $+0.14$ | $+0.13$ | $+0.15$ | 1.01 | $+0.12$ | $+0.11$ |
| $n_s$ | $+0.83$ | $+0.81$ | $+0.83$ | 0.99 | $+0.03$ | $+0.03$ |
| $\ln(10^{10}A_s)$ | $+0.34$ | $+0.35$ | $+0.37$ | 1.00 | $+0.12$ | $+0.10$ |
| $\omega_m$ | $+0.20$ | $+0.22$ | $+0.21$ | 1.00 | $-0.03$ | $-0.03$ |

- The shift does not depend on which code made the mock (agreement to $\leq 0.03\sigma$): it is a fixed offset between the two models.
- The linear Fisher forecast (CosmoPower model on a CLASS mock, deck 1) predicted every shift to within $0.03\sigma$.
- Self-fits are unbiased, so the MCMC machinery is fine; the offset is the emulator's accuracy at the damping tail.

Pull = (mean − input)/$\sigma$. Numbers in `data/bench_mocks.json`, made by `scripts/fig_bench_mocks.py`.

# Benchmark with the newer emulator: smaller code bias




Cross fits only: each code fitted to the other code's mock (ΛCDM, one bin, same input cosmology). Top: CP_paper networks (Spurio Mancini et al. 2022) vs CLASS with massless $\nu$. Bottom: CosmoPower `lcdm v1` networks, Bolliet et al. 2023 (arXiv:2303.01591), vs CLASS with the same physics ($1 \times 0.06$ eV $\nu$, HMcode lensing).

Largest pull drops from $0.86\sigma$ ($n_s$) to $0.27\sigma$; $\omega_b$ $0.60 \to 0.16\sigma$, $A_s$ $0.46 \to 0.23\sigma$, $\omega_m$ $0.25 \to 0.09\sigma$. Errors unchanged.

# The SPT emulator: what it is and how close to CLASS

CosmoPower `SPT_high_accuracy` ΛCDM networks, Balkenhol et al. (SPT-3G 2018 TT/TE/EE), trained on high-precision CAMB.

- Inputs $\omega_b, \omega_c, h, \tau, n_s, \ln(10^{10} A_s)$; lensed TT, TE, EE for $\ell = 2$–$5000$, in $\mu\mathrm{K}^2$. **No lensing ($\phi\phi$) network.**
- Training box (inferred): $\omega_c \in [0.090, 0.160]$, $h \in [0.55, 0.80]$, $\tau \leq 0.090 \Rightarrow$ each $\omega_m$ bin must stay in $\sim [0.112, 0.182]$.
- Physics: one massive $\nu$ (0.06 eV) and HMcode lensing, as `lcdm v1`.

| | CP_paper | `lcdm v1` (Bolliet+23) | `SPT_high_accuracy` |
|---|---|---|---|
| trained on | CAMB, massless $\nu$ | CLASS, $m_\nu$, HMcode | CAMB, $m_\nu$, HMcode |
| $C_\ell$ vs matching CLASS ($\ell \leq 2508$) | TT $-0.7$%, EE $-1$% (tail) | $\sim 0.1$% TT, EE | 0.06% TT, 0.08% EE rms |
| $\ell$ range | 2–2508 | 2–11 000 | 2–5000 |
| $\phi\phi$ | yes | yes | no |
| parameter bias | $\leq 0.86\sigma$ (MCMC) | $\leq 0.27\sigma$ (MCMC) | $\leq 0.17\sigma$ (Fisher) |

Bias: cross fits of the emulator model to a CLASS mock (TT, TE, EE + lensing, Planck-like noise). SPT: Fisher forecast, TT/TE/EE from SPT; with `lcdm v1`'s $\phi\phi$ added the forecast bias is $\leq 0.2\sigma$.

# The three emulators: inputs and training ranges

| | CP_paper<br>Spurio Mancini+22 | `lcdm v1`<br>Bolliet+23 | `SPT_high_accuracy`<br>Balkenhol+ | our priors<br>(`with_cp_lcdm`) |
|---|---|---|---|---|
| $\omega_b$ | [0.005, 0.040] | [0.0193, 0.0253] | [0.018, 0.026] | [0.0193, 0.0253] |
| $\omega_c$ | [0.001, 0.99] | [0.0799, 0.200] | [0.090, 0.160] | via $\omega_m$ |
| $h$ | [0.20, 1.00] | $H_0 \in$ [40.05, 100] | [0.55, 0.80] | [0.60, 0.75] |
| $\tau$ | [0.01, 0.80] | [0.020, 0.120] | [0.018, 0.090] | [0.02, 0.10] |
| $n_s$ | [0.70, 1.30] | [0.881, 1.049] | [0.868, 1.130] | [0.92, 1.00] |
| $\ln(10^{10}A_s)$ | [1.61, 5.00] | [2.50, 3.50] | [2.70, 3.20] | [2.90, 3.20] |
| $\Rightarrow \omega_m$ per bin | no real limit | $\approx$ [0.103, 0.223] | $\approx$ [0.112, 0.182] | [0.11, 0.21] |
| fixed physics | massless $\nu$*, lin. lensing* | $1 \times 0.06\,\mathrm{eV}$ $\nu$, HMcode | $1 \times 0.06\,\mathrm{eV}$ $\nu$, HMcode | |
| training code | CAMB | CLASS | CAMB (high accuracy) | |
| outputs | TT, TE, EE, $\phi\phi$ | TT, TE, EE, $\phi\phi$ | TT, TE, EE | |
| $\ell$ range | 2–2508 | 2–11 000 | 2–5000 | |
| input names | `omega_b, omega_cdm, h, tau_reio, n_s, ln10^{10}A_s` | `omega_b, omega_cdm, H0, tau_reio, n_s, ln10^{10}A_s` | `ombh2, omch2, h, tau, ns, logA` | |

Same six ΛCDM parameters in all three; none has $m_\nu$, $N_{\rm eff}$ or $Y_{\rm He}$ as an input (fixed in training). `lcdm v1` $\phi\phi$ has no $\tau$ input. CP_paper ranges as coded in our stand-in (full training box of the paper); `lcdm v1` and SPT ranges inferred from the stored parameter mean/std, assuming uniform sampling. *Not stated in the paper; inferred from our comparison with CAMB/CLASS (best match). With SPT, our $\tau$ prior (to 0.10) and $\omega_m$ prior (to 0.21) would need tightening.

# Estimating $\partial C_\ell / \partial \omega_m$ at fixed $\theta_s$

**CLASS, numerical** (benchmark): five-point stencil, $\epsilon = 1.2 \times 10^{-3}$,

$$\frac{\partial C_\ell}{\partial \omega_m} \simeq \frac{C_\ell(-2\epsilon) - 8C_\ell(-\epsilon) + 8C_\ell(\epsilon) - C_\ell(2\epsilon)}{12\,\epsilon},$$

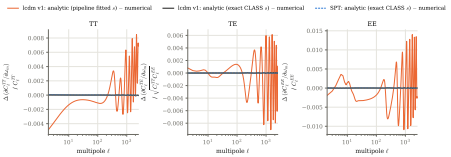
with $h$ re-solved at every point so that CLASS's $100\theta_s = 1.0419693546$ (achieved to $4 \times 10^{-9}$). High precision: $\ell_{\rm max} = 3500$, refined $\ell$ and $q$ sampling, tight tolerances ($\sim$9 min/run).

**Emulators, analytic**: exact network Jacobian (no step), along the fixed-$\theta_s$ direction

$$\left.\frac{\partial C_\ell}{\partial \omega_m}\right|_{\theta_s} = \left.\frac{\partial C_\ell}{\partial \omega_c}\right|_{h} + \left.\frac{\partial C_\ell}{\partial h}\right|_{\omega_c} \left.\frac{dh}{d\omega_m}\right|_{\theta_s}, \qquad \left.\frac{dh}{d\omega_m}\right|_{\theta_s} = -3.524$$

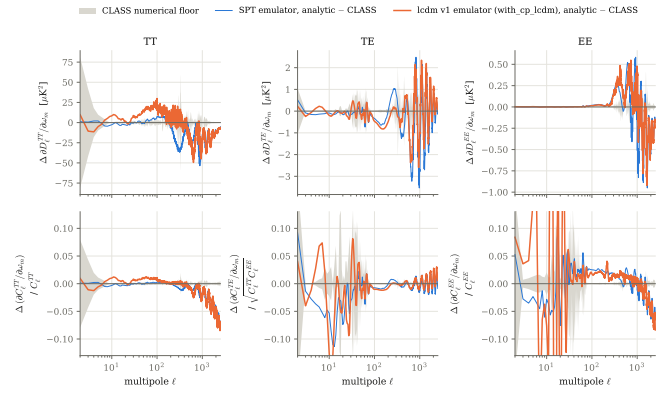
(slope from CLASS; $d \ln h / d \ln \omega_m = -0.748$).

Fiducial: $\omega_b$ 0.02237, $\omega_c$ 0.12, $h$ 0.674, $\tau$ 0.054, $n_s$ 0.965, $\ln(10^{10} A_s)$ 3.044, $m_\nu$ 0.06 eV, HMcode.




Analytic − numerical, same emulator, per unit $\omega_m$, / $C_\ell$. With the exact slope: $\lesssim 10^{-4}$ (Jacobians exact). The pipeline's fitted slope ($-0.74844$ vs $-0.74773$) alone moves the `lcdm v1` derivative by up to 0.8% (TT), 1.4% (EE).

# $\partial C_\ell / \partial \omega_m$: emulators vs CLASS




Top: absolute ($\mu$K$^2$ per unit $\omega_m$). Bottom: per unit $\omega_m$, relative to $C_\ell$ (TE: $\sqrt{C^{TT} C^{EE}}$). Grey: CLASS numerical floor.

| $\sqrt{\chi^2}$, $1\sigma$ shift | TT | TE | EE |
|---|---|---|---|
| template itself | 2.04 | 1.92 | 3.04 |
| SPT − CLASS | 0.071 | 0.026 | 0.063 |
| `lcdm v1` − CLASS | 0.080 | 0.030 | 0.067 |
| CLASS floor | 0.005 | 0.006 | 0.009 |

$\delta\omega_m = 0.0012$; cosmic variance only, $f_{\rm sky} = 0.57$, $30 \le \ell \le 2508$.

Both emulators reproduce the CLASS derivative to 1.5–4%, SPT slightly better. The error grows with $\ell$ with the same shape for both: TT $-4\%$ of $C_\ell$ per unit $\omega_m$ at $\ell = 1500$–$2000$, $-7$ to $-8\%$ at 2500.

# A smooth $\omega_m(\ell)$: the parametrisation

Instead of $\omega_m$ bins with sharp edges, a smooth $\ell$-dependence added to the spectra in linear response:

$$C_\ell = C_\ell^{\Lambda\mathrm{CDM}}(\omega_b, h, \tau, n_s, A_s, \omega_{m0}) + \left.\frac{\partial C_\ell}{\partial \omega_m}\right|_{\theta_s,\,\omega_b,\,\tau,\,n_s,\,A_s} \delta\omega_m(\ell), \qquad \delta\omega_m(\ell) = a + b\,x + c\,x^2, \quad x = \frac{\ell - \ell_c}{1000}.$$

- Same $\delta\omega_m(\ell)$ on TT, TE, EE, $\phi\phi$, each with its own derivative. The derivative is the exact network Jacobian (`lcdm v1`) at the *current* point, along the fixed-$\theta_s$ direction: $\partial_{\omega_c} C_\ell|_h + \partial_h C_\ell\,(dh/d\omega_m)_{\theta_s}$, with $d\ln h/d\ln\omega_m|_{\theta_s}$ from a fit to CLASS (good to 0.3% over the prior box).
- $a$ is degenerate with $\omega_{m0}$ (plus the matching $h$ shift): $a \equiv 0$, so $\omega_{m0} = \omega_m(\ell_c)$ and $b$, $c$ carry all the $\ell$-dependence.
- $\ell_c = 400$: where $\omega_{m0}$ and $b$ decorrelate (Fisher). Moving $\ell_c$ only reparametrises the same model.
- Fisher: $\sigma(b) = 0.0041$, $\sigma(c) = 0.0053$, $\sigma(\omega_{m0}) = 0.0013$. Priors $b, c \in [-0.05, 0.05]$ (flat).
- $\sigma[\omega_m(\ell)]$: 0.0014 at $\ell \simeq 200$–$400$, 0.0027 at 1200, 0.005 at 1600, 0.018 at 2500: the data constrain the curve only below $\ell \sim 1500$.

Linear response is accurate near $b = c = 0$ (error $\sqrt{\chi^2} = 0.01$ for a $1\sigma$ shift, 0.1 for $3\sigma$), i.e. for a null test. Far out in the high-$\ell$ tails it is a phenomenological model, not "$\omega_m$ at that $\ell$".

# Mock vs model: the runs

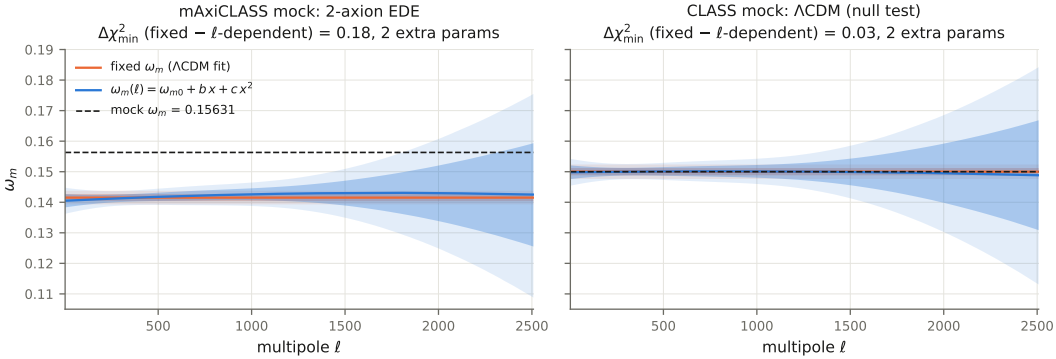
Two mocks, each fitted with fixed $\omega_m$ ($\Lambda$CDM) and with $\omega_m(\ell)$. Sampler: CosmoPower `lcdm v1`; likelihood `fake_planck_realistic` (TT, TE, EE + lensing, Planck-like noise, $\ell \leq 2508$, noiseless mocks). 8 chains $\times 10^5$ steps, burn-in 0.3.

| mock | model | $\chi^2_{\rm min}$ | acceptance | max $R-1$ | extra params |
|---|---|---|---|---|---|
| CLASS $\Lambda$CDM | fixed $\omega_m$ (`one_bin`) | 0.11 | 0.28 | 0.0005 | – |
| | $\omega_m(\ell)$ (`poly`) | 0.08 | 0.26 | 0.0011 | $b$, $c$ |
| mAxiCLASS EDE | fixed $\omega_m$ (`one_bin`) | 5.38 | 0.27 | 0.0003 | – |
| | $\omega_m(\ell)$ (`poly`) | 5.20 | 0.25 | 0.0004 | $b$, $c$ |

- **CLASS $\Lambda$CDM mock**: $\omega_b$ 0.0224, $h$ 0.674, $\tau$ 0.054, $n_s$ 0.965, $\ln(10^{10}A_s)$ 3.044, $\omega_m$ 0.15; $1 \times 0.06$ eV $\nu$, HMcode.
- **mAxiCLASS mock**: 2-axion EDE ($n = 3$), Bella, Poulin, Vagnozzi, Knox 2026 PDH0 posterior means: $f_{\rm EDE} = 0.085$, 0.073 at $\log_{10} a_c = -3.44$, $-4.00$; $H_0 = 72.6$, $\omega_m = 0.1563$, $\omega_b$ 0.02267, $n_s$ 0.988, $\tau$ 0.060, $\ln(10^{10}A_s)$ 3.067; same $\nu$ and lensing.

One `one_bin` chain on the $\Lambda$CDM mock stopped at $7.4 \times 10^4$ steps. Numbers in `data/poly_runs.json`.

# Can $\omega_m(\ell)$ see the EDE in the mock?




Bands: 68/95%. Orange: fixed $\omega_m$ (flat). Blue: $\omega_m(\ell) = \omega_{m0} + b\,x + c\,x^2$, $x = (\ell - 400)/1000$. Dashed: the mock's $\omega_m$.

**No.** On the EDE mock $b = 0.0023 \pm 0.0039$, $c = -0.0009 \pm 0.0050$ (joint $\chi^2 = 0.44$ vs 0, 2 dof); $\Delta\chi^2_{\rm min} = 0.18$. The null test behaves the same ($b, c$ within $0.1\sigma$, $\Delta\chi^2_{\rm min} = 0.03$). The $\omega_m(\ell)$ curve stays flat; both fits sit 12–13$\sigma$ below the mock's $\omega_m$.

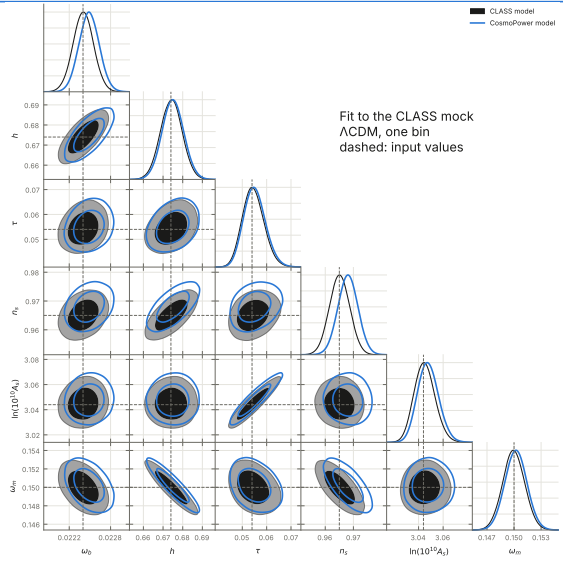
# EDE mock: ΛCDM absorbs the EDE into parameter shifts

| | mock | fixed $\omega_m$ | | $\omega_m(\ell)$ | |
|---|---|---|---|---|---|
| | | mean $\pm\sigma$ | pull | mean $\pm\sigma$ | pull |
| $\omega_b$ | 0.02267 | $0.02221 \pm 0.00015$ | $-3.1$ | $0.02216 \pm 0.00018$ | $-2.8$ |
| $h$ | 0.7265 | $0.6338 \pm 0.0054$ | $-17.1$ | $0.6329 \pm 0.0061$ | $-15.2$ |
| $\tau$ | 0.060 | $0.0596 \pm 0.0045$ | $-0.1$ | $0.0597 \pm 0.0045$ | $-0.1$ |
| $n_s$ | 0.988 | $0.9682 \pm 0.0036$ | $-5.5$ | $0.9692 \pm 0.0044$ | $-4.2$ |
| $\ln(10^{10}A_s)$ | 3.067 | $3.0501 \pm 0.0087$ | $-2.0$ | $3.0507 \pm 0.0087$ | $-1.9$ |
| $\omega_{m0}$ | 0.1563 | $0.1415 \pm 0.0011$ | $-12.9$ | $0.1416 \pm 0.0013$ | $-11.8$ |
| $b$ | – | – | | $0.0023 \pm 0.0039$ | $+0.6$ |
| $c$ | – | – | | $-0.0009 \pm 0.0050$ | $-0.2$ |
| $\chi^2_{\min}$ | 0 | 5.38 | | 5.20 | |

Pull = (mean − mock)/$\sigma$; for $b$, $c$ vs 0. The mock's "$\omega_m$" is the total matter density of the EDE cosmology: a ΛCDM fit is not expected to recover it.

At Planck-like noise this EDE is almost a ΛCDM cosmology with different parameters ($H_0$ 72.6 → 63.4, $\omega_m$ 0.156 → 0.142, $n_s$ 0.988 → 0.968). The residual it leaves ($\chi^2_{\min} = 5.4$) does not have the shape of a smooth $\delta\omega_m(\ell)$: $b$, $c$ remove only 0.18 of it, and the other parameters barely move.

# Backup: full posteriors, CLASS mock



# Backup: full posteriors, CosmoPower mock

